IoT-Based LPG Gas Leakage Detection and Safety Automation System Using ESP32

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Abstract

Liquefied Petroleum Gas (LPG) is widely used in residential, commercial, and industrial applications. However, gas leakage can lead to fire hazards, explosions, and health risks. This paper presents an IoT-based LPG Gas Leakage Detection and Safety Automation System using the ESP32 microcontroller and MQ gas sensor. The proposed system continuously monitors LPG concentration levels and categorizes the environment into multiple safety levels. Upon detection of hazardous gas concentrations, the system automatically activates a ventilation fan, shuts off the gas valve using a servo motor, and sends emergency notifications to users through mobile applications and SMS alerts. The system also uploads real-time sensor data to the ThingSpeak cloud platform for remote monitoring and analysis. Experimental results demonstrate the effectiveness of the system in providing rapid response and enhanced safety.

Keywords: LPG Leakage Detection, ESP32, MQ Sensor, IoT, ThingSpeak, Safety Automation, Servo Motor.

I. Introduction

Liquefied Petroleum Gas (LPG) is one of the most commonly used fuels for cooking and industrial applications due to its high calorific value and clean-burning characteristics. Despite its advantages, LPG leakage remains a major safety concern. Gas leaks may occur due to damaged pipelines, faulty regulators, or improper handling of cylinders.

Traditional gas leakage detection systems mainly provide local alarms without implementing automatic preventive actions. Recent advancements in the Internet of Things (IoT) have enabled intelligent monitoring systems capable of real-time data acquisition, cloud connectivity, and automated emergency responses.

This paper proposes an IoT-based LPG Gas Leakage Detection and Safety Automation System that not only detects gas leakage but also performs automatic safety actions such as shutting off the gas supply, activating ventilation mechanisms, and notifying users remotely.

II. Literature Survey

Several LPG leakage detection systems have been proposed in previous studies. Most systems utilize gas sensors and microcontrollers to detect leakage and trigger alarms. However, many existing solutions lack cloud monitoring capabilities and automatic safety mechanisms.

Recent IoT-based systems have improved monitoring capabilities through cloud integration. However, limitations such as delayed response, absence of automatic gas shut off, and limited risk classification still exist. The proposed system addresses these shortcomings by integrating real-time cloud monitoring, multi-level risk assessment, automatic gas valve control, and emergency notification features.

III. Proposed System

The proposed system consists of:

- ESP32 Microcontroller
- MQ Gas Sensor
- Servo Motor
- Exhaust Fan
- Buzzer
- ThingSpeak Cloud Platform
- GSM/Mobile Notification Module

Working Principle

1. The MQ sensor continuously monitors LPG concentration.
2. The ESP32 reads sensor values and compares them with predefined thresholds.
3. Based on gas concentration levels, the system classifies the environment into:
    a. Safe Level
    b. Warning Level
    c. Danger Level
    d. Critical Level
4. During warning conditions, visual and audible alerts are generated.
5. During danger conditions:
    a. Ventilation fan is activated.
    b. Emergency notifications are sent.
6. During critical conditions:
    a. The server motor closes the gas valve.
    b. Fan remains activated.
    c. Continuous emergency alerts are issued.
7. Sensor readings are uploaded to ThingSpeak for real-time monitoring.

IV. Hardware Components

A. ESP32 Microcontroller

ESP32 serves as the central controller of the system. It processes sensor data, controls actuators, and enables Wi-Fi communication with cloud platforms.

B. MQ Gas Sensor

The MQ sensor detects LPG concentration in the surrounding environment and converts gas concentration into analog voltage signals.

C. Servo Motor

The servo motor is used for automatic gas valve shut-off during critical leakage conditions.

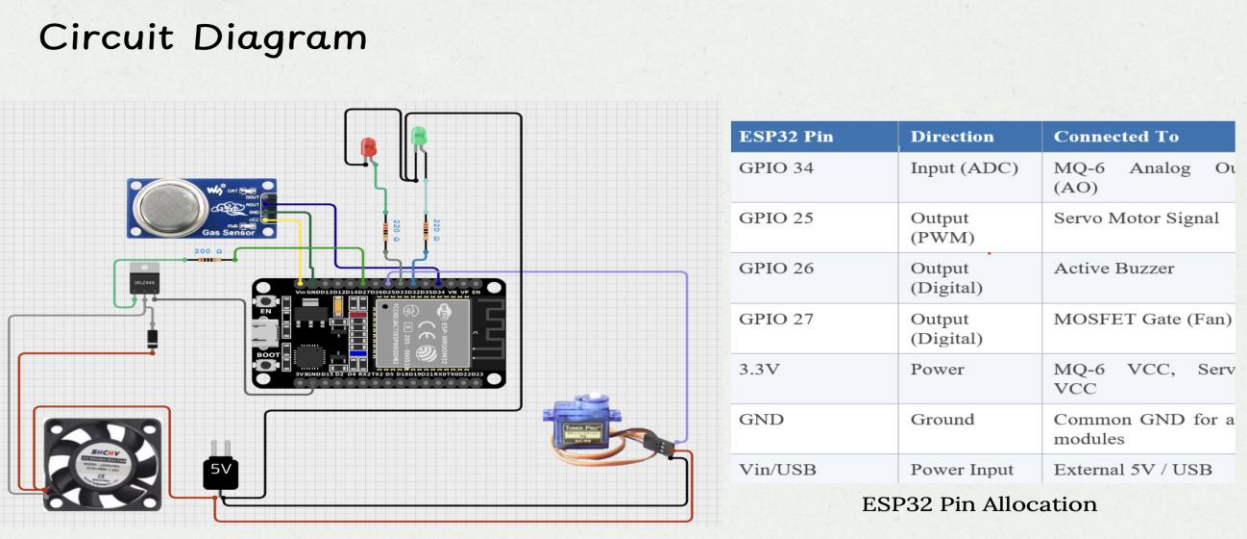
D. Ventilation Fan

The fan helps remove accumulated gas from the environment, reducing explosion risks.

E. Buzzer

The buzzer provides immediate local alerts whenever leakage is detected.

V. Circuit Diagram



ESP32 Pin	Direction	Connected To
GPIO 34	Input (ADC)	MQ-6 Analog Ou (AO)
GPIO 25	Output (PWM)	Servo Motor Signal
GPIO 26	Output (Digital)	Active Buzzer
GPIO 27	Output (Digital)	MOSFET Gate (Fan)
3.3V	Power	MQ-6 VCC, Serv VCC
GND	Ground	Common GND for a modules
Vin/USB	Power Input	External 5V / USB

ESP32 Pin Allocation

VI. System Architecture

The architecture of the proposed system includes gas sensing, signal processing, cloud communication, and safety automation modules.

Gas Leakage → MQ Sensor → ESP32 Controller → Decision Logic

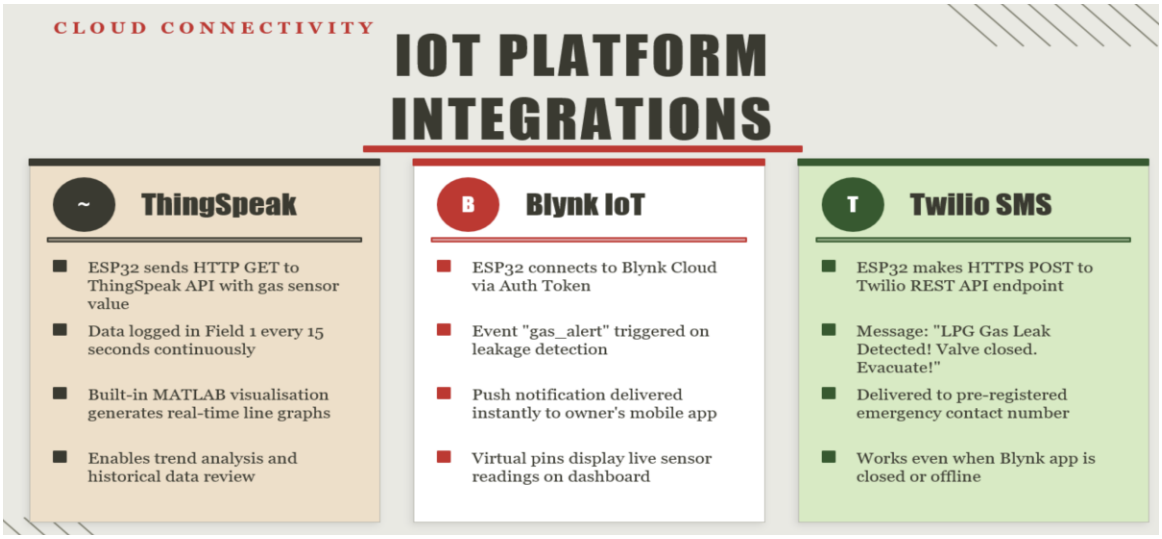
→ Buzzer Alert

→ Ventilation Fan Control

→ Servo Motor Gas Shut-Off

→ ThingSpeak Cloud Monitoring

→ SMS/Mobile Notifications

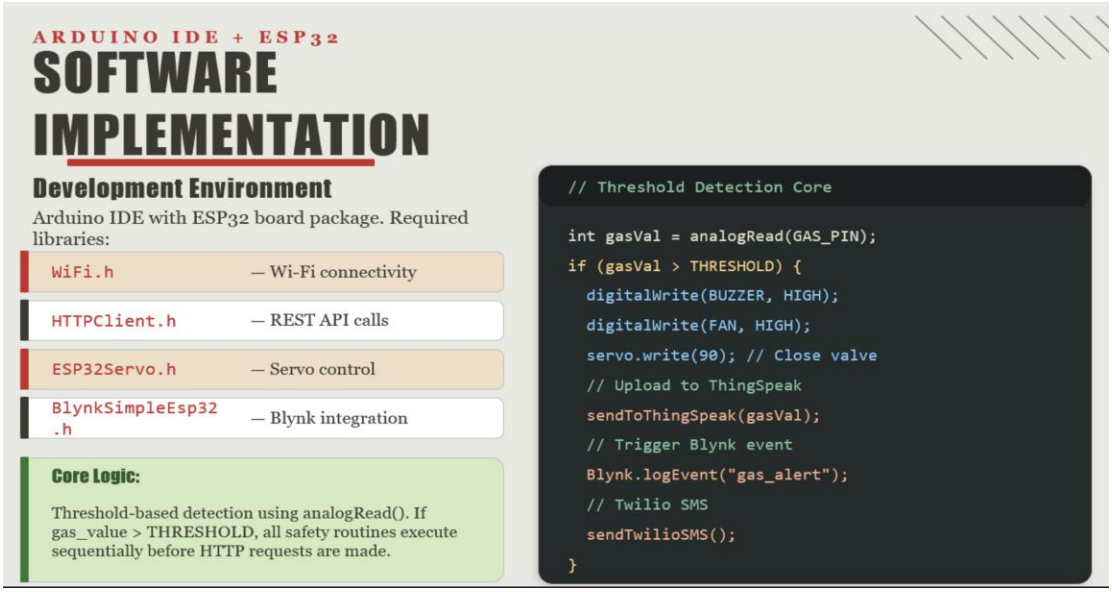


VII. Software Implementation

The system software is developed using Arduino IDE.

Algorithm

1. Initialize ESP32 and sensor modules.
2. Read MQ sensor value.
3. Compare sensor value with threshold levels.
4. If gas concentration exceeds warning level:
    a. Activate buzzer.
5. If gas concentration exceeds danger level:
    a. Start ventilation fan.
    b. Send notification.
6. If gas concentration exceeds critical level:
    a. Close gas valve using servo motor.
7. Upload sensor data to ThingSpeak.
8. Repeat continuously.



VIII. Experimental Results

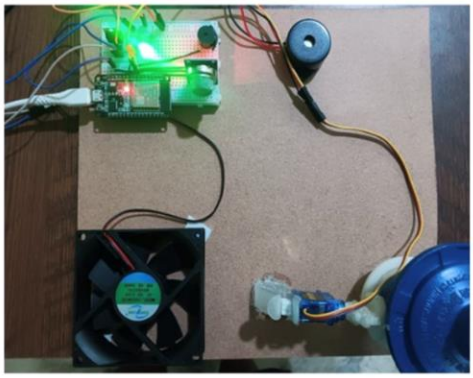
The system was tested under various LPG concentration levels.

Gas Level	Sensor Reading	System Action
Safe	Low	Monitoring Only
Warning	Medium	Buzzer Activated
Danger	High	Fan Activated + Notification
Critical	Very High	Gas Valve Closed + Emergency Alert

The results indicate that the proposed system successfully detects LPG leakage and performs safety actions with minimal response time. Real-time cloud monitoring through ThingSpeak enables users to remotely observe gas concentration trends.

No spurious actuations occurred. MQTT reporting was uninterrupted throughout, with status "SAFE" confirmed on the mobile dashboard.

The clean-air baseline of 285 ± 42 ADC counts on the MQ-6 (compared to a typical MQ-2 baseline of 380 ± 90 ADC counts in the same environment) confirms the MQ-6's more stable response in ambient air — a consequence of its narrower gas-sensitivity profile reducing the response to background volatile organic compounds.



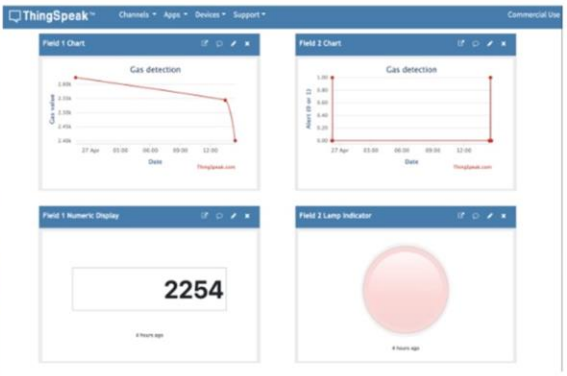


Figure 6.1: Serial Monitor Output — SAFE State (ADC readings, 500 ms interval)

IX. Advantages

- Real-time gas leakage detection.
- Automatic gas valve shut-off.
- Cloud-based monitoring.
- Mobile notifications and alerts.
- Reduced risk of fire and explosions.
- Low-cost implementation.
- Easy installation and maintenance.

X. Applications

- Residential kitchens.
- Restaurants and hotels.
- Industrial gas storage facilities.
- Laboratories.
- Commercial buildings.

XI. Future Scope

Future enhancements may include:

- AI-based leakage prediction.
- Integration with smart home systems.

- Battery backup during power failures.
- Mobile application with live analytics.
- Multi-gas detection capability.

XII. Conclusion

This paper presented an IoT-based LPG Gas Leakage Detection and Safety Automation System using ESP32 and MQ gas sensors. The system effectively detects gas leakage, performs automatic safety actions, and provides real-time cloud monitoring. The integration of servo-controlled gas shut-off, ventilation control, and emergency notifications significantly improves safety and response time. Experimental results demonstrate that the proposed system is reliable, cost-effective, and suitable for residential and industrial applications.

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